KK-L-156b
July 29, 1963
SUPERSEDING
Int. Fed. Spec. KK-L-00156a (Army-GL)
January 31, 1963 and
Fed. Spec. KK-L-156
May 16, 1956
(see 6.5)

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, HARNESS, VEGETABLE-TANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers harness leather made from cattlehides and primarily used for harness straps.

1.2 Classification. The harness leather shall be furnished in the following types, classes, and selections as specified (see 6.2):

Type I.—Sides.
Type II.—Backs.
Class 1.—Mildew-resistant treated.
Sub-class a.—0.18 to 0.60 percent fungicide.
Sub-class b.—0.18 to 0.70 percent fungicide.
Class 2.—Untreated.
Selection A.
Selection B.
Selection C.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

KK-L-311—Leather; Methods of Sampling and Testing.
QQ-S-781—Steel Strapping, Flat.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
CCC-C-429—Cloth, Cotton, Osnaburg.
CCC-C-467—Cloth, Jute (or Kenaf), Burlap.
PPP-B-636—Box, Fiberboard.
PPP-P-291—Paperboard, Wrapping, Cushioning.
PPP-T-60—Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents,

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U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-T-40625 — Tubing, Bias Sewn (Burlap or Osnaburg) Cloth.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society For Testing and Materials Standards:

D 1814—1962—Measuring Thickness Of Leather Units

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 **Material.** The leather shall be cattlehides (see 6.4).

3.2 **Tannage.** The leather shall be tanned by a vegetable-tanning process.

3.3 **Finish.** The leather shall be full grain or full grain lightly snuffed. The flesh side shall be smooth and free from flesh, and may be buffed, shaved, or skived to secure greater uniformity of thickness.

3.4 **Trim (form) and area.** The trim of the sides or backs shall be as specified by the definitions for "sides" and "backs" in section 5 of KK-L-311. The area expressed in terms of square feet shall be legibly marked on the butt area of the flesh side of the leather.

3.5 **Color.** The color of the harness leather shall be as specified (see 6.2).

3.6 **Selections.**

3.6.1 *Selection A.* Each side or back of leather shall be free from deep cuts, damaged grain, open scratches, grubs, brands, hip marks, and rough shoulders, or any other defects which affect cutting value.

3.6.2 *Selection B.* Each side or back of leather may contain imperfections not permitted in selection A, but such imperfection or combination of imperfections, shall not show a loss in cutting value greater than 6 percent of the total area of the side or back.

3.6.3 *Selection C.* Each side or back of leather may contain imperfections not permitted in selection A, but such imperfection or combination of imperfections shall not show a loss in cutting value greater than 12 percent of the total area of the side or back.

3.7 **Physical requirements.**

3.7.1 *Weight and thickness.* The leather shall be of the weight and thickness as specified (see 6.2) and as shown in table I.

TABLE I.—*Thickness*

Weight	*Thickness (64ths inch)*
Extra light	8 and below 10
Light	10 and below 12
Medium	12 and below 14
Heavy	14 and below 16
Extra heavy	16 and above

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3.7.2 *Breaking force.* Not less than 80 percent of the specimens tested shall have breaking-force values of not less than that specified in table II when tested as specified in 4.3.

TABLE II.—*Breaking force*

Thickness (inches)	*Breaking force (pounds)*
8/64 and 9/64	135
10/64 and 11/64	170
12/64 and 13/64	210
14/64 and 15/64	240
16/64 and above	260

3.7.3 *Elongation.* Not less than 80 percent of the specimens tested shall show elongation values of 20.0 percent or less when under load stresses as specified in table III when tested as specified in 4.3.

TABLE III.—*Elongation*

Thickness (inches)	*Load stress (pounds)*
8/64 and 9/64	110
10/64 and 11/64	140
12/64 and 13/64	170
14/64 and 15/64	200
16/64 and above	230

3.7.4 *Cracking.* The leather shall not crack when tested as specified in 4.3.

3.8 **Chemical requirements.**

3.8.1 *Class* 1. When class 1 leather is specified, the leather shall be treated with the fungicide paranitrophenol (see 6.3). Sub-class "a" leather shall contain 0.18 to 0.60 percent fungicide based upon the dry weight of the leather and sub-class "b" leather shall contain 0.18 to 0.70 percent fungicide based upon the dry weight of the leather when tested as specified in 4.3.

3.8.2 The leather shall conform to the chemical requirements specified in table IV, when tested as specified in 4.3.

TABLE IV.—*Chemical requirements*

Characteristics	Minimum	Maximum
Chloroform extract[1] (percent)	15.0	35.0
Water soluble materials[1] (percent)	—	18.0
Total ash[1] (percent)	—	5.0
Degree of tannage[1] (percent)	50.0	75.0
Acidity (pH)	3.3	4.5

[1] Calculated to moisture free basis.

3.9 **Workmanship.** The finished harness leather shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection for acceptance.** Inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Examination of the end item.* The classification of defects found during the examination shall be in accordance with the list shown in 4.2.1.1 and 4.2.1.2. The sample unit for these examinations shall be one side or back. The lot size shall be expressed in terms of one side or back. The inspection level and acceptable quality level for these examinations shall be in accordance with 4.2.1.3.

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4.2.1.1 *Examination for visual characteristics.*

Examine	Defect
Trim	Not as specified, i.e., not back or side.
Color	Not as specified.
Finish	Not full grain or full grain slightly snuffed. Flesh side not smooth or contains areas of loose flesh in excess of 4 square inches in total area.
Quality of leather:	
Selection A	Deep cuts, open scratches, damaged grain, open grub holes, brands, hip marks, rough shoulders or other defects which affect cutting value.
Selection B	Any defect or combination of defects listed for selection A, impairing cutting value of more than 6 percent of the area of the side or back.
Selection C	Any defect or combination of defects listed for selection A, impairing cutting value of more than 12 percent of the area of the side or back. NOTE.—Light, well-healed scratches or grubs, light fat wrinkles or slight salt stains shall not be classified as defects.

4.2.1.2 *Examination for dimensional characteristics.*

Examine	Defect
Thickness	Thickness of leather not as specified. NOTE.—Thickness shall be measured by ASTM Designation D 1814.
Area (expressed in terms of square feet)	Less than area marked on the side or back.

4.2.1.3 *Inspection levels and acceptable quality levels (AQL's).* The inspection levels and acceptable quality levels expressed in defects per hundred units shall be as follows:

Examination	Inspection level	AQL's
For defects applicable to 4.2.1.1, visual characteristics	II	4.0
For defects applicable to 4.2.1.2, dimensional characteristics	L-6	4.0

4.2.2 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete wrapping and adequate securing of rolls, incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling. Bulging or distortion of containers. Closure of cloth-covered bundles incorrect, insecurely sewn bundle affecting serviceability.
Content	Not rolled grain out in set of 3 sides or 6 for backs, and the final piece of each roll not rolled flesh side out.
Weight	Net weight exceeds requirement.

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TABLE V.—*Tests*

Characteristic	Requirement paragraph	Test methods	Requirement applicable to	
			Sample unit	Lot average
Material identification[1]	3.1	Standard Commercial	X	—
Tannage[1]	3.2	Standard Commercial	X	—
Breaking force	3.7.2	4.3.1	X	—
Elongation	3.7.3	2021[2]	X	—
Cracking	3.7.4	4.3.2	X	—
Paranitrophenol[3]	3.8.1	4.3.3	—	X
Chloroform extract[3]	3.8.2	6311	—	X
Water-soluble materials[3]	3.8.2	6341	—	X
Total ash[3]	3.8.2	6421	—	X
Degree of tannage[3]	3.8.2	6641	—	X
Acidity (pH)	3.8.2	6621	—	X

[1] The contractor shall submit a certificate of compliance for these characteristics.

[2] Exception to method 2021—The elongation of the sample shall be recorded to the nearest 1.0 percent.

[3] Calculated to moisture free basis.

4.3 End item testing. The methods of testing specified in KK-L-311 wherever applicable and as listed in table V shall be followed. Sampling procedures and location from which the sample unit is to be obtained shall be in accordance with section 6 of KK-L-311. The specimens for testing-strength requirements shall be selected from the corner of the test sample unit which is closest to the first "A" in the word "AREA" of figure 6, section 6 of KK-L-311. When chemical requirements are specified on a moisture-free basis, the method for moisture determination shall be method 6221 of KK-L-311. When chemical requirements are specified on a hide-substance basis, the method for hide-substance determination shall be method 6461 of KK-L-311. The lot shall be rejected if any one of the following conditions exist:

(*a*) More than three test failures occur for either the breaking force or elongation requirement.

(*b*) More than one test failure occurs for the remaining requirements applicable to the sample unit.

(*c*) Any lot average value which fails to meet the specified requirement.

4.3.1 *Breaking force.* Method 2011 of KK-L-311 shall be used except that the term "breaking force" shall be used in lieu of "tensile strength" where applicable in this test method. The scope shall be revised to state that this method is intended for use in determining the breaking-force value of leather. The cross-sectional area of the specimen and the calculation of tensile strength need not be determined. The breaking force of the sample shall be recorded to the nearest pound.

4.3.2 *Cracking.* Method 4011 of KK-L-311 shall be used except that the results shall be reported as "pass or fail". The diameter of the mandrel used shall be as follows:

Weight of leather	Diameter of mandrel
Extra light	5/16 inch
Light	3/8 inch
Medium	7/16 inch
Heavy	1/2 inch
Extra heavy	9/16 inch

4.3.3 *Test for the determination of paranitrophenol.* Method 6711 of KK-L-311 shall be used except that two ml. of either

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chloroform extract shall be transferred to the ninth flask containing sodium citrate (20 ml. of a 10 percent solution of $Na_3C_6H_5O_7$ in water). If the concentration of the developed paranitrophenol color is too intense for instrument readings, the 2 ml. of aliquot shall be suitably reduced. To compute the percent paranitrophenol on a dry basis use the following formula:

$$\text{Percent paranitrophenol (dry basis)} = \frac{\text{Milligrams paranitrophenol in aliquot} \times \text{Dilution}}{10\times \text{Dry weight of specimen in grams}}$$

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A or B.* Leather of one type, selection, weight, class, and color only shall be neatly and tightly rolled together in multiple sets of three for sides and six for backs. When a roll of leather consists of more than one set, each additional set shall be alternated so that the heads of one set are adjacent to the butts of the next set. The leather shall be rolled grain side out except that the outer piece shall be rolled flesh side out. Each roll of leather shall be completely wrapped in 80-pound-minimum-basis weight kraft paper conforming to grade B of UU-P-268. The paper shall be folded to inclose each end of the roll and shall be secured with gummed kraft paper tape. Alternatively, each roll of leather shall be completely wrapped in single-faced corrugated fiberboard conforming to type II of PPP-P-291. The wrap shall be folded over to completely inclose each end of the roll. The wrap shall be secured in place by tying with jute or cotton cord or twine, applied twice around the body, approximately 12 inches from each end (when applicable) and once lengthwise, tying with nonslip knots.

5.1.2 *Level C.* Leather shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Each roll of leather, packaged as specified in 5.1, shall be completely inclosed in waterproofed kraft paper conforming to class B-2 of UU-P-271. The paper inclosing each end shall be tightly folded in place so that the roll can be inserted into cloth covering material. The waterproofed roll of leather shall be baled in bias-sewn tubing or a bag (presewn one end) made from cotton-osnaburg cloth conforming to type I, class 2, finish A of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The bias-sewn tubing specified shall conform to MIL-T-40625. Bales shall have the open ends closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, of 0.072-inch-thick galvanized soft iron or steel wire with a 1/2-inch-diameter formed eye at each end. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Each bale shall be reinforced with two flat steel straps conforming to type I, class A, 5/8 inch wide by 0.015 inch thick of QQ-S-781. The straps shall be applied around the bale placed approximately 10 inches from each end. Sufficient tension shall be applied to the strapping, insuring tight application. Alternatively, each waterproofed roll shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style FOL of PPP-B-636. All seams, corners, and manufacturer's joints shall be sealed by the application of 3-inch-minimum width olive-drab color

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cloth tape conforming to type III, class 1 of PPP-T-60. Each fiberboard shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Weight of the contents of each bale shall not exceed 100 pounds with a 10-pound plus tolerance permitted, and for fiberboard shipping containers the weight limitations shall be in accordance with the requirements of the container specification.

5.2.2 *Level B.* Each roll of leather, packaged as specified in 5.1, shall be baled in bias-sewn tubing or a bag (presewn one end) made from cotton-osnaburg cloth conforming to type I, class 2, finish A of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The bias-sewn tubing specified shall conform to MIL-T-40625. Bales shall have the open ends closed with two double-looped wire ties. The wire ties shall be not less than 6 inches long, of 0.072-inch-thick galvanized soft iron or steel wire with a 1/2-inch-diameter formed eye at each end. The first wire tie shall be applied as close to the bale as possible so there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being 3 to 5 inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately 1 inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions. Each bale shall be reinforced with two flat steel straps conforming to type I, class A, 5/8 inch wide by 0.015 inch thick of QQ-S-781. The straps shall be applied around the bale approximately 10 inches from each end. Sufficient tension shall be applied to the strapping insuring tight application. Alternatively, each roll shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style OSC (overlap not less than 3 inches) of PPP-B-636. Each fiberboard shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. Weight of the contents of each bale shall not exceed 100 pounds with a 10-pound plus tolerance permitted, and for fiberboard shipping containers the weight limitations shall be in accordance with the requirements of the container specification.

5.2.3 *Level C.* Leather, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The leather is intended for use in safety belts and other general harness strap applications.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and selection required (1.2).
(*c*) Color required (3.5).
(*d*) Weight and thickness required (3.7.1).
(*e*) Selection of applicable levels of packaging and packing (5.1 and 5.2)

6.3 For class 1 leather to be used in direct contact with the skin, sub-class "a"

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(0.18 to 0.60 percent fungicide) should be used. For class 1 leather not used in direct contact with the skin, sub-class "b" (0.18 to 0.70 percent fungicide) should be used.

6.4 Bull hides are excluded from the cattlehide to be used for harness since they do not produce good-quality leather. A bull hide usually ranges in weight from 75 to more than 100 pounds in the green, salted state. It is characterized by very thick and rough neck and shoulders.

6.5 This specification is a revision of Federal Specification KK-L-156 dated May 16, 1956. The comparison of types and classes is as follows:

KK-L-156b	*KK-L-156*
Types	*Form (trim)*
Type I Sides	Sides
Type II Backs	Backs
Classes	*Types*
Class I	Type I—Treated for resistance to mildew
Sub-class a	—
Sub-class b	—
Class 2	Type II—Untreated.

6.6 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Leather, cattlehide, in the original tannery shapes.
Carload minimum weight 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Leather, cattlehide, in the original tannery shapes.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTERESTS:

Army—GL

Navy—S

Air Force—ASD

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